

YUEZHEN (LILY) DONG

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RELEVANT SKILLS

Languages	Python, Java, TypeScript/JavaScript, C/C++, SQL, Ruby, Bash, Go
ML/AI	PyTorch, scikit-learn, Gemini API, RAG, Embeddings, LLM Agents, GRPO, NeMo-RL
Frameworks	FastAPI, Flask, React, JavaFX, Node/Express, Ruby on Rails, Spring Boot, GraphQL
Infrastructure	Docker, PostgreSQL, AWS, Git, Linux, CI/CD, monitoring, distributed training

EDUCATION

University of Waterloo - Faculty of Engineering	Expected Graduation MAY 2028
BASc in Computer Engineering – President’s Scholarship of Distinction	Waterloo, ON, CAN

ACHIEVEMENTS

IBDP Graduate CaYPT (Canadian Young Physicists’ Tournament) National Champion IYPT (International Young Physicists’ Tournament) Bronze Medalist FIRST Robotics Provincial Semi-Finalist

RESEARCH EXPERIENCE

Grätzel Cell JUL 2023 - JUL 2024
Python Matplotlib, COMSOL Multiphysics, Multimeter, Lux Meter

- Constructed and optimized a **dye-sensitized solar cell** using titanium dioxide and iodine electrolyte to study the mechanisms of **photovoltaic energy conversion**.
- Evaluated the influence of **electrolyte concentration**, glass conductivity, TiO_2 layer thickness, and temperature on cell efficiency through electrochemical analysis and precise measurements using **multimeters** and **lux meters**.
- Employed **MATLAB** for data modeling and simulations to assess the impact of parameter variations on electron transport and photocurrent, validating experimental results with theoretical predictions.

Faraday Waves JUL 2023 - JUL 2024
MATLAB, ImageJ, Tracker, Python, Matplotlib

- Investigated **Faraday waves** and oscillating droplets, examining surface waves and instabilities in vertically oscillating systems.
- Analyzed surface and **gravity-capillary waves** using **ImageJ** and **Tracker** to study the effects of frequency, amplitude, and viscosity on wave patterns.
- Simulated droplet dynamics in **MATLAB** and processed data using **Python** and **Matplotlib** to quantify variations and validate theoretical predictions.

Magnetostriction SEP 2022 - JUL 2023
MATLAB, ImageJ, Tracker, Python, Matplotlib

- Investigated the impact of external **magnetic fields** on magnetization and magnetostriction in various materials, focusing on **ferromagnetic**, **paramagnetic**, and **diamagnetic** properties.
- Measured **B-H curves** using a custom-built coil and H-bridge circuit to control magnetic fields, with real-time data captured by a **Hall effect sensor** to analyze magnetic saturation and hysteresis effects.
- Quantified magnetostriction in ferrite rods using a **strain gauge** and **Wheatstone Bridge** setup, capturing precise dimensional changes under varying magnetic fields.

Saffman–Taylor Instability of Miscible Fluids JUL 2022 - JUL 2023
ImageJ, Tracker, MATLAB, Python, Matplotlib

- Conducted research, data analysis, and experiments on three-stage Marangoni flow and **Saffman–Taylor instability**, focusing on fluid dynamics and fractal patterns.
- Analyzed fractal dimensions and geometry using **ImageJ** and **Tracker** to develop a systematic theory on the role of viscosity in Saffman–Taylor instability.
- Simulated fluid behavior with **MATLAB** and processed data using **Python** and **Matplotlib** to quantify viscosity changes in paint samples with different concentrations.

Acoustic Analysis of Airflow-Induced Sound in Perforated Rotating Disks JUL 2022 - JUL 2023
MATLAB, ImageJ, Tracker, Python, Matplotlib

- Investigated the sound phenomena of airflow interacting with a **rotating disk with holes**, analyzing how sound characteristics change with parameters like flow rate, rotational speed, and hole configuration.

- Developed experimental setups using a **motor speed controller**, air compressor, and microphone to capture sound data, with real-time analysis and visualization conducted in **MATLAB** and **Python**.
- Quantified relationships between **sound intensity**, frequency, and disk parameters, using theoretical models to validate experimental observations and identify key parameter impacts.

Pulsating Heat Tube in a Condensing-Heating-Condensing (CHC) System

JUL 2022 - JUL 2023

Arduino, EasyEDA, OnShape, Python, JSON, OpenCV, Tracker, Matplotlib

- Researched thermodynamics and fluid dynamics of pulsating heat tubes to analyze **two-phase heat transfer efficiency**.
- Developed a Python-based tool using **OpenCV Object Detection** to extract and analyze thermocouple data from 8 channels for real-time monitoring.
- Simulated vapor motion within the heat tube using Python, achieving a **96% match** between simulations and experimental results.

Marangoni Flow and Rayleigh-Taylor Instability in Evaporating Fluids

JUL 2021 - JUL 2022

Arduino, EasyEDA, ImageJ, Tracker, Matplotlib, Comsol Multiphysics, MATLAB

- Investigated Marangoni flow and Rayleigh-Taylor instability by analyzing fluid dynamics at interfaces to study **droplet behavior** in experiments.
- Designed and implemented a droplet release system using **Arduino** and **3D-printed components** to enhance precision and reproducibility in experiments.
- Developed and optimized an interferometry analysis system with **OpenCV** for high-resolution tension measurement and analysis.
- Performed dynamic simulations using MATLAB and Comsol, achieving **93% accuracy** to verify experimental results with simulations.

WORK EXPERIENCE

Meta Production Engineering Fellow

Jun 2026 – Sep 2026

Linux, Python, Bash, Docker, CI/CD, Networking, Monitoring, Site Reliability Engineering

MLH Fellowship

- Built Linux-based reliability tooling for deployment readiness checks, service health monitoring, and log-driven diagnostics, automating routine triage workflows with **Python**, **Bash**, and Dockerized test environments.
- Developed CI/CD validation scripts and failure-analysis reports covering networking failures, resource saturation, configuration drift, and dependency issues, improving the handoff from alert detection to root-cause investigation.

Software Engineering Intern

May 2026 – Aug 2026

Python, TypeScript, React Native, Ruby on Rails, RAG, LLM, GRPO, OAuth, BLE

Shopify

- Raised Sidekick Analytics subagent unit-test pass rate from a **43.4% baseline to 87%** by building an end-to-end agentic RL pipeline with **GRPO/NeMo-RL**, converting production-failure tests into verifiable rewards and scoring grouped rollouts across held-out tests, tool errors, and judge metrics.
- Implemented **RAG-based context retrieval** with embedding reranking and prompt orchestration pipelines for agentic AI tooling infrastructure, reducing irrelevant context fed to the LLM by **~40%** and improving AI-generated code acceptance rate in developer review.
- Built **Quick Device Activation** for Shopify POS using **BLE** pairing, QR/manual-code fallback, and **OAuth Device Authorization** – reducing merchant device setup time from **~5 minutes to under 30 seconds**.

Software Engineering Intern

Sep 2025 – Dec 2025

React, TypeScript, Ruby on Rails, GraphQL

Shopify

- Owned a **staff assignment modal** in Retail Admin end-to-end – built React components and designed **GraphQL** queries with server-side pagination, reducing staff search query response time by **~60%**.
- Shipped full-stack features across **6 systems** – Admin Web, POS Mobile, Shopify Core, Business Platform, Cloud Sync Streamer, and POS Channel – maintaining backward-compatible data contracts and achieving **zero regression**.
- Coordinated with engineering, product, and design to clarify requirements and document tradeoffs, reducing mid-sprint scope changes and contributing to on-time delivery of **3 consecutive merchant-facing feature releases**.

Web Developer Co-op

Jan 2025 – Apr 2025

Java, Spring Boot, Maven, Apache POI, Git

AGF Investments

- Built a **Java/Spring Boot** ScoreCard application with XLSX streaming via **Apache POI**, configurable transformation pipelines, and input validation, automating report generation across **4 departments** and saving **20+ staff hours/week** previously spent on manual spreadsheet processing.